Polynomial-time simulation of non-Clifford quantum error correction

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arXiv

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Overview

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1 **Most magic-state preparation protocols**

Circuits made of CNOT, T, Paulis, Z measurements and *compatible* X measurements.

Examples:

- Magic state distillation and cultivation based on applying a logical 3rd level non-Clifford gate to a code.
- Code switching between 3D and 2D codes.

Overview

Most magic-state preparation protocols can be simulated in polynomial-time

1. Most magic-state preparation protocols
2. **Simulated**

Maintain state representation throughout the circuit.
Two representations: Schrödinger (phase polynomials) and Heisenberg (generalized stabilizer group).

Can compute the following:

- Pauli expectation values and amplitudes
- Measurement samples
- State equality

Overview

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1. Most magic-state preparation protocols
2. Simulated
3. **Polynomial time**

Polynomial-time complexity in the number of qubits and circuit depth
No exponential in the number of T gates!

Outline

1. The phase polynomial representation
2. Compatible measurements
3. DCP stabilizer formalism
4. Benchmarking our simulator

1 The phase polynomial representation

Phase polynomial states

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Question: what happens when we apply G_3 gates to $|0\rangle^{\otimes n} |+\rangle^{\otimes m}$?

$$U |0\rangle^{\otimes n} |+\rangle^{\otimes m} = \sum_{y \in \mathbb{F}_2^m} U |x\rangle$$

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New question: what is $U |x\rangle$ for $U \in G_3$?

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$$T_i\,|x\rangle = \omega^{x_i}\,|x\rangle$$

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for a so-called **third-order phase polynomial**:

$$f(x) = \sum_i a_i x_i + 2\sum_{i<j} b_{ij} x_i x_j + 4\sum_{i<j<k} c_{ijk} x_i x_j x_k$$

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with S an affine subspace of $\mathbb{F}_2^{n+m}$ and q a third-order phase polynomial.

Phase polynomial states

Definition (3PP state): a third-order phase polynomial (3PP) state is a state of the form

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Question: but what about measurements?

2 Compatible measurements

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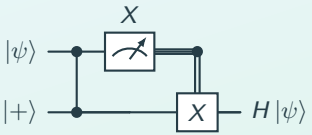
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$\Rightarrow$ Still a 3PP state!

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Observation 1: not every X measurement can be compatible.

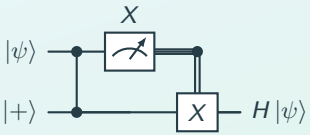
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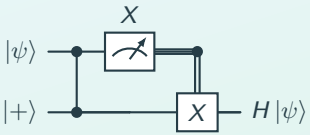


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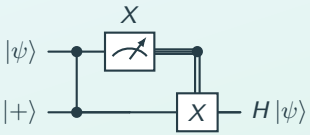
$$H|0\rangle = |+\rangle \quad \text{(compatible)}, \qquad H|T\rangle \propto \cos\tfrac{\pi}{8}|0\rangle - i\sin\tfrac{\pi}{8}|1\rangle \quad \text{(incompatible)}.$$

The latter has unequal amplitude magnitudes, so it is not a 3PP state.

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Question: which X measurements are compatible with a state $|\psi\rangle$ specified by $(S, q(x))$?

Compatible X measurements: support and phase

Claim: compatibility of measuring $X(v)$ is determined by whether $S = S + v$ and, if so, by the form of $\Delta_v q(x) = q(x+v) - q(x) \pmod 8$ on S.

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If $S = S + v$: relabeling the second sum gives

$$\begin{aligned} \Pi_\mu |\psi\rangle &\propto \sum_{x \in S} \omega^{q(x)} + (-1)^\mu \omega^{q(x+v)} |x\rangle \\ &\propto \sum_{x \in S} \omega^{q(x)} \left(1 + \omega^{\Delta_v q(x) + 4\mu}\right) |x\rangle \end{aligned}$$

Classification of compatible X measurements

Theorem: measuring $X(v)$ on a 3PP state (S, q) is compatible iff

1. $S \neq S + v$, or
2. $S = S + v$ and one of the following holds on S (modulo 8):

(2a) $\Delta_v q(x) = 4f(x)$, f affine,

(2b) $\Delta_v q(x) = 2 + 4g(x)$, $2g$ a third-order phase polynomial,

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Proof: $X(v)$ shifts the support, so $S + v = S$. Matching amplitudes at $x \in S$ gives

$$\omega^{q(x+v)} = (-1)^{\mu}\omega^{q(x)} \;\Rightarrow\; q(x+v) = q(x) + 4\mu \pmod 8 \;\Rightarrow\; \Delta_v q(x) = 4\mu \pmod 8.$$

Thus $\Delta_v q = 4f$ with $f(x) = \mu$ constant, hence affine: case (2a).

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Proof: for $x \in S$, $Z(w)\,|\psi\rangle = |\psi\rangle$ gives $w \cdot x = 0$. Anticommutation gives $w \cdot v = 1$. Thus:

$$w \cdot (x + v) = 0 + 1 \neq 0 \pmod 2 \;\Rightarrow\; x + v \notin S \;\Rightarrow\; S \neq S + v.$$

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Noise model: DCP errors at every circuit location

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Theorem (measurement compatibility under noise): assume

1. Circuit made of CNOT-dihedral gates and compatible measurements
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Examples:

- **Distillation and non-Clifford computation**: logical CNOT-dihedral gates on CSS codes, followed by ideally deterministic checks.
- **DCP checks in cultivation**: controlled-DCP gates give deterministic ancilla X readouts.
- **CSS gauge fixing and code switching**: new X checks are fixed or anticommute with a current Z constraint.

3 DCP stabilizer formalism

DCP stabilizer formalism

Question: can we come up with a stabilizer picture for 3PP states?

Observation: if we start with a state stabilized by a Pauli operator S, and apply a gate $U \in G_3$, then the new state is stabilized by $S' = USU^\dagger$, **which is a DCP operator**.
$\Rightarrow$ Let's look at states stabilized by a group of DCP operators. We call these **DCP states**.

Example: $|00\rangle$ is stabilized by $\mathcal{S}_{\text{DCP}} = \langle S_1, S_2, \text{CZ}_{12} \rangle$. Interestingly:

1. It has $m = 3$ independent generators, but $n = 2$ qubits. Contrasts with Pauli case, where $n = m$.
2. It is also uniquely stabilized by some dim-2 subgroups of $\mathcal{S}_{\text{DCP}}$: $\langle Z_1, Z_2 \rangle$, $\langle Z_1, S_2 \rangle$,$\langle S_1, Z_2 \rangle$, etc.

Questions:

1. Is every DCP state uniquely stabilized by an n-dimensional DCP subgroup?
2. What is the link between DCP states and 3PP states?
3. Can we build a *DCP tableau simulator*, with update rules for compatible measurements?

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The product state $|\phi\rangle$ is uniquely specified by n Pauli stabilizers $P_1, \ldots, P_n$. Conjugate them through the preparation circuit:

$$g_i = UP_iU^\dagger, \qquad g_i\,|\psi\rangle = UP_i\,|\phi\rangle = |\psi\rangle\,.$$

From phase polynomials to stabilizers

Theorem: DCP states are exactly the 3PP states.

First direction: every 3PP state is a DCP state. By Proposition 1, it can be prepared as

$$|\psi\rangle = U\,|\phi\rangle\,, \qquad |\phi\rangle = |0\rangle^{\otimes(n-r)}\,|+\rangle^{\otimes r}\,, \quad U \in G_3.$$

The product state $|\phi\rangle$ is uniquely specified by n Pauli stabilizers $P_1, \ldots, P_n$. Conjugate them through the preparation circuit:

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Answer to the first question: n independent commuting DCP generators suffice.

From stabilizers to support and phases

Let G uniquely fix $|\psi\rangle$. Write its elements as $DX(v)$, with D diagonal Clifford.

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Combine the quadratic Clifford phases along a path from x_0. Each binary choice of shift adds one factor, giving a third-order phase polynomial. $\Rightarrow$ A 3PP state!

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Unitary gates: conjugate each generator. For $U \in G_3$, the result is still DCP:

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Result: compatibility tests and tableau updates take polynomial time.

4 Benchmarking our simulator

MERLIN: from theory to benchmarks

MERLIN: our open-source simulator for non-Clifford QEC

Input: Stim-like circuit descriptions.

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Benchmark circuits:

- Bravyi–Haah distillation: $3k + 8 \to k$ magic states, increasing k.
- 15-to-1 distillation and distance-3 / distance-5 cultivation.
- CCZ code switching with $[\![27, 3, 3]\!]$ and $[\![81, 3, 5]\!]$ codes[1].

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Comparison: Tsim[2], Xtim[3], Clifft[4] and SymFT[5].

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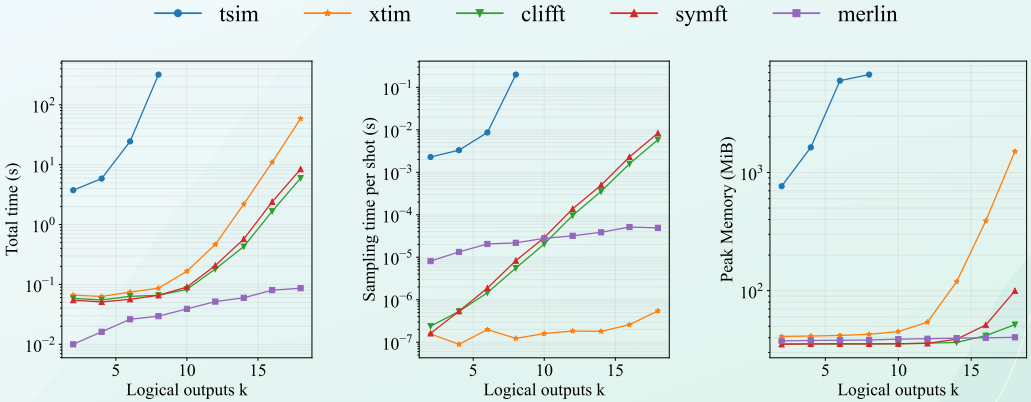
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Metrics: total runtime, sampling time per shot and peak memory.

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Bravyi–Haah: scaling with logical outputs



1000 attempted shots, $p = 10^{-3}$.

Total includes pre-processing. Sampling time excludes pre-processing.

Distillation, cultivation and code switching

Total runtime in ms, including pre-processing

Circuit	Tsim	Xtim	Clifft	SymFT	**MERLIN**
15-to-1	5029.95	85.86	99.56	57.67	**12.52**
Cultiv-3	OOM	65.29	54.29	63.63	**28.98**
Cultiv-5	OOM	253.51	**67.93**	577.20	202.01
Switch-3	OOM	2817.53	OOM	ERR	**573.69**
Switch-5	OOM	OOM	OOM	ERR	**6959.13**

Time per shot in μs, excluding pre-processing

Circuit	Tsim	Xtim	Clifft	SymFT	**MERLIN**
15-to-1	3446.85	0.40	0.37	**0.18**	9.62
Cultiv-3	OOM	0.39	0.51	**0.34**	28.34
Cultiv-5	OOM	7.70	7.97	**6.52**	200.15
Switch-3	OOM	**2.65**	OOM	ERR	549.93
Switch-5	OOM	OOM	OOM	ERR	**6704.94**

1000 attempted shots, $p = 10^{-3}$. Cultivation excludes the escape stage.
OOM = allocation above 20 GB. ERR = runtime error. Cultiv / Switch suffix = distance.

Future work

1. There are still a few circuits that evade our simulation method (e.g. H-based cultivation, some more complicated distillation protocols, etc.). Can we extend our method to cover these cases?
2. Can we formalize a notion of **"DCP detector error model"** for non-Clifford circuits?
3. Can we build a theory of **"DCP codes"**, generalizing XP codes?
4. Can we construct an interesting **resource theory** based on 3PP states? A "DCP polytope" generalizing the stabilizer polytope?
5. Is there any application to **learning theory**?
6. What concrete non-Clifford protocols should we start comparing and benchmarking using Merlin?

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